

1. Purpose

The resources available in Microsoft Azure are finite. Microsoft is continually adding new resources; however, demand occasionally exceeds the available resources.

This document discusses some of the capacity related topics in Microsoft Azure.

2. Table of Contents

1.	Purpose	1
2.	Table of Contents	1
3.	Reference	1
3.1.	Azure Resource Availability	1
3.2.	Azure Infrastructure	1
3.3.	Quotas	2
3.4.	Reserved Instances	2
3.5.	Capacity Reservations	2
3.6.	Differences Between Capacity Reservations and Reserved Instances	3
3.7.	Resource Allocation	3
3.8.	TD SYNEX Support	3
3.9.	Rental Car Analogy	4
4.	Associated Documents	4
5.	Document Approval	4
6.	Document Revision History	4

3. Reference

3.1. Azure Resource Availability

Microsoft Cloud Operations is a dedicated capacity team that forecasts future requirements to ensure necessary capacity is structured and available for customer and internal use.

All service teams at Microsoft include capacity planning as a key feature of their datacenter models and data replication plans to ensure that there is necessary capacity for information processing, telecommunications, and environmental support.

Planning and forecasting do not always keep up with spikes in demand. Adding resources to a data center does not happen instantly. Space must be allocated, infrastructure may need to be added or upgraded, hardware must be ordered, assembled and installed, and new systems must be tested. All of this takes significant time.

Other than for capacity reservations, Microsoft offers no SLAs or assurances that capacity will be available when a VM is started.

3.2. Azure Infrastructure

Azure delivers virtual machines through software, but every deployment still depends on physical infrastructure built in data centers with finite servers, power, cooling, and space. When demand spikes, the platform must allocate real hardware in a specific region, and if that capacity is already in use, requests fail because no additional compute exists at that moment.

The idea of unlimited cloud resources is a misconception; scaling requires building new facilities, installing racks, and provisioning equipment, a process that takes quarters or years. This challenge

is compounded by global supply constraints, including shortages of CPUs, memory, and other components, along with long manufacturing and delivery lead times for server hardware.

As demand for cloud and AI workloads continues to outpace the speed of infrastructure expansion, capacity limits remain a normal and expected part of operating at hyperscale.

3.3. Quotas

Many Azure services have quotas, which are the maximum number of resources for your Azure subscription. Each quota represents a specific countable resource, such as the number of virtual machines you can create, the number of storage accounts you can use concurrently, the number of networking resources you can consume, or the number of API calls to a particular service you can make.

The concept of quotas is designed to help protect customers from things like inaccurately resourced deployments and mistaken consumption. For Azure, it helps minimize risks from deceptive or inappropriate consumption and unexpected demand. Quotas are set and enforced in the scope of the subscription.

Having an available quota does not equate to resources being available when needed. A quota only allows a resource to be provisioned when there are resources available in the data center.

Only TD SYNEX can request an increase in quota for a subscription. There is no guarantee that a request will be approved by Microsoft. It is the customer's responsibility to monitor quota utilization and request additional quota to meet projected demands. Quota requests can take days for Microsoft to approve or deny.

Careful consideration should be given to any request for quota increases. As previously discussed, quotas help protect from inappropriate consumption which can include fraudulent usage. Requesting excessive quota can expose the customer to high fraud charges and Microsoft will not credit for fraud. Fraud charges are the responsibility of the reseller to pay and hopefully collect from the end user.

3.4. Reserved Instances

Azure Reservations help you save money by committing to one-year or three-year plans for multiple products. Committing allows you to get a discount on the resources you use.

After purchase, the reservation discount automatically applies to the resource usage that matches the attributes you select when you buy the reservation.

Reserved Instances are only a discounting mechanism. There is no guarantee that the resource will be available. For this reason, reserved instances are usually purchased for machines that run 24x7.

Reserved instances for virtual machines offer an optimization setting called "capacity priority". This setting does not guarantee that resources will be available when needed. It only prioritizes available capacity. The only way to ensure that capacity will be available is through Capacity Reservations.

3.5. Capacity Reservations

On-demand Capacity Reservation enables you to reserve Compute capacity in an Azure region or an Availability Zone for any duration of time.

Once you create the Capacity Reservation, the resources can be used immediately. Capacity is reserved for you until you delete the reservation.

Capacity Reservations are priced at the same rate as the underlying VM size. Both used and unused Capacity Reservation are eligible for Saving Plan and Reserved Instances term commitment discounts. Only the compute component is charged for unused Capacity Reservations.

3.6. Differences Between Capacity Reservations and Reserved Instances

Differences	On-demand Capacity Reservation	Reserved Instances
Term	No term commitment required. Can be created and deleted as per the customer requirement	Fixed term commitment of either one-year or three-years
Billing discount	Charged at pay-as-you-go rates for the underlying VM size*	Significant cost savings over pay-as-you-go rates
Capacity SLA	Provides capacity guarantee in the specified location (region or availability zone)	Doesn't provide a capacity guarantee. Customers can choose "capacity priority" to gain better access, but that option doesn't carry an SLA
Region vs Availability Zones	Can be deployed per region or per availability zone	Only available at regional level

3.7. Resource Allocation

Creating a virtual machine in Azure does not allocate resources. Resources are only allocated when the machine is in a running or stopped state. The state of the machine can be determined in the Azure portal.

Stopped State

The stopped state is when you are logged in to the operating system of an Azure VM, you can issue a command to shut down the server. If you find the VM in the Azure console, you'll see the state listed as "Stopped". The hardware is still allocated to the VM and the VM is being billed.

Deallocated State

If you stop the virtual machine through Azure, rather than through the OS, it goes into a "Stopped (deallocated)" state. The hardware is released and is available for other customers. The VM compute charge is no longer charged.

Since a stopped VM does not release the hardware, you can be certain that the machine will restart. When restarting from a deallocated state, if no hardware is available, the machine will not start.

3.8. TD SYNEX Support

Microsoft does not coordinate capacity allocation decisions with TD SYNEX, and TD SYNEX does not receive a reserved pool of Azure capacity to distribute to customers.

If a customer cannot deploy a virtual machine due to capacity constraints, they should contact TD SYNEX support at CSPSupport@tdsynnex.com. The support team will engage Microsoft to confirm current availability and, when possible, provide an estimated timeline for capacity, though this estimate is not guaranteed.

TD SYNEX can request escalation on behalf of the customer, but escalation does not create capacity or accelerate availability. Escalation has not created new capacity in the past attempts.

TD SYNEX leadership maintains ongoing discussions with Microsoft regarding capacity constraints, and Microsoft continues to expand infrastructure as quickly as possible, within the limits of hardware supply, data center space, and broader industry constraints affecting all hyperscale providers.

3.9. Rental Car Analogy

When renting a car, the car is reserved for your exclusive use (allocated VM resource). The rental car company is not able to rent it to another party because you have it. You pay the rental car company for the car whether you use it or not.

If you park the car in your driveway and do not use it (stopped VM), the rental company is still not able to rent it to another party; therefore, you continue to pay the company for the car.

If you return the car to the company (deallocated VM), they can then rent it to another party. You are not obliged to pay for the car even if it is sitting on the company's lot for an extended period. You may decide you need the car again at some point; however, if it were rented to another party and no other cars are available, you will not get a car (allocation failure).

It is possible, you could ask for another size of car, perhaps a mid-size instead of the compact you wanted and be able to get a car (changing VM family). This is not an automatic process.

You can commit to renting a car of a certain size from the company for the next year (reserved instance). You will pay for the car at a discounted rate, whether you use it or not. Since you will pay for it even if you do not use it, it would make sense to keep it in your driveway (allocated resource) even if you do not use it. If you choose to not take possession of it, there is no guarantee that a car of that size will be available at the rental company lot when you need it.

Keeping the car in your driveway has additional costs. You may need to pay the daily LDW fee (OS cost). You have the option to pay just the rental cost (compute cost) and have the rental company hold a car on its lot until you need it (capacity reservation). Although there is a cost, it is lower than taking possession of the car but still guarantees it will be available when you need it.

If a meteor hits the card while held on the lot for you and makes it unavailable to you when you need it, the rental car company will reimburse you for the time that it is not available to you when you request it (SLA for Capacity Reservation).

4. Associated Documents

- [Datacenter architecture and infrastructure - Microsoft Service Assurance | Microsoft Learn](#)
- [Troubleshooting Azure VM allocation failures - Virtual Machines | Microsoft Learn](#)
- [Quotas overview - Azure Quotas | Microsoft Learn](#)
- [On-demand Capacity Reservation in Azure - Azure Virtual Machines | Microsoft Learn](#)
- [What are Azure Reservations? - Microsoft Cost Management | Microsoft Learn](#)
- [Service Level Agreement for Online Services \(microsoft.com\)](#)

5. Document Approval

This document has been approved by:

- Manager, Cloud Sales Engineering

6. Document Revision History

Number	Effective Date	Revision History	Author
1.0	11/28/2023	Initial Draft	David Bloom
1.1	6/11/2026	Added TD SYNEX support	David Bloom

Azure Virtual Machine Capacity



Type:	7: Reference	Version:	1.1
Status:	Final	Effective:	6/11/2026